Vehicle Orientation Estimation Using Conventional Image Processing

Ahmedabad University

Group 1 | Project 8 | Mid-Semester Presentation

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Digital Image Processing (Monsoon 2026)

Motivation

- Drones are now used for **traffic monitoring** in Indian cities.
- Knowing the **direction of each vehicle** helps in lane flow study, parking analysis and wrong-way detection.
- Deep models need GPUs and huge labelled data. **Classical methods** are light, fast and easy to explain.
- Aerial vehicles are **tiny (often < 30 px)**, parked close, and covered by shadows or trees.
- Goal: see how far plain **DIP tools** (edges, PCA, Hough) can go.



Source: DRASHTI-HaOBB, Zenodo (2026)

Problem Definition

Problem

- Given an aerial vehicle crop, estimate its **body axis angle** using only conventional image processing, with no learning.

Target (heading θ → undirected axis φ)

$$r = \theta \bmod 180°, \quad \varphi = \min(r, 180° - r) \in [0°, 90°]$$

Method A: PCA on Canny edge pixels

$$C = (1/N) \Sigma (p_i - \mu)(p_i - \mu)^T, \quad axis = v(\lambda_{max})$$

Method B: Hough lines (length weighted)

$$\varphi_p = \tfrac{1}{2} \cdot atan2(\Sigma w_i \sin 2\alpha_i, \Sigma w_i \cos 2\alpha_i)$$

Error

$$e = |\varphi_p - \varphi| \ \text{in degrees}$$

Literature Review

Work	Key idea	What we took
Zhou et al. (2017)	Canny + Radon transform on aerial cars	Edge based axis estimate
Rybski et al. (2010)	HOG features for vehicle orientation	Feature option for later
Hu (1962)	2nd order moments give shape orientation	Base of our PCA step
Duda & Hart (1972), Matas et al. (2000)	Hough and probabilistic Hough lines	Method B
Canny (1986), Zuiderveld (1994)	Edge detector, CLAHE contrast	Preprocessing
Mardia & Jupp (2000)	Mean of axial data using 2α	Hough angle fusion
Xia et al. (2018), DOTA	Oriented boxes for aerial objects	Reading OBB labels

- **Gap:** most work uses learning or clean scenes; little on tiny vehicles in dense Indian traffic.

Results (Initial Run)

Validation split: 40 images, 1822 crops (difficulty 0)

Metric	PCA	Hough
Coverage (%)	99.89	99.84
MAE (°)	11.15	14.103
Median error (°)	6.282	6.819
P90 error (°)	27.212	39.133
Within 10° (%)	65.22	62.67

- PCA has **lower error**.
- Both fail on **crops < 20 px** (MAE > 30°).
- Shadows and road lines pull the axis away.



Fig: Sample predictions on validation crops

Next Plan

- **Method C:** Otsu mask → minimum-area rectangle and image moments.
- $\tan 2\varphi = 2\mu_{11} / (\mu_{20} - \mu_{02})$
- **Ablation:** Canny thresholds, CLAHE on/off, blur size.
- **Front vs back:** intensity asymmetry along the axis to get full 0° to 360° heading.
- **Scale up:** full val split, then train_original labels for tuning.
- **Error analysis:** by crop size, shadow and occlusion.

Week	Target
4 to 5	Method C + ablation
6 to 7	Heading (front/back)
8	Full evaluation, final report

Individual Contribution

Member	Contribution
Vehan Gajjar	Label parser, crop extraction with 10% padding, metadata.csv, dataset notes
Vedant Solanki	Preprocessing (blur, CLAHE, Canny) and PCA estimator with rejection rules
Abhi Jain	Hough line estimator, doubled angle fusion, method comparison notes
Krish Prajapati	Angle folding + unit tests, evaluation metrics, overlay plots
All members	Literature review (Weeks 1 to 3), weekly reports, slides

THANK YOU